
Confidence Calibration under Ambiguous Ground Truth

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Abstract

1 Confidence calibration assumes a unique ground-truth label for each input, yet
2 many real-world tasks—medical imaging, natural language inference, autonomous
3 perception—exhibit irreducible label ambiguity where rational annotators gen-
4 uinely disagree. We identify a systematic *calibration gap*: when standard post-hoc
5 calibrators (Temperature Scaling, Platt scaling, histogram binning) are fitted on
6 majority-voted labels, they appear well-calibrated under hard-label evaluation
7 while remaining significantly overconfident relative to the true annotator distri-
8 bution. This failure is universal across parametric and non-parametric methods
9 because all share the same voted-label training target—not an insufficiency in
10 model capacity. This gap is the direct calibration analogue of the coverage gap in
11 conformal prediction under ambiguous ground truth, and we prove its direction is
12 determined by the majority-vote bias. To close the gap, we propose three post-hoc
13 recalibration methods—**Soft-Label Temperature Scaling (SLTS)**, **Monte Carlo**
14 **Temperature Scaling (MCTS)**, and soft-label non-parametric calibration—that
15 replace hard-label targets with annotator distributions and optimize proper scoring
16 rules. On CIFAR-10H (51 human annotations per image), our methods reduce
17 soft ECE by up to 63% relative to TS while maintaining competitive hard-label
18 calibration. On a skin-disease benchmark (ISIC 2019, 8 classes, $K=9$ annotators),
19 the calibration gap under TS reaches +9.6 pp, which SLTS reduces to +0.5 pp.

20 1 Introduction

21 A well-calibrated classifier $f : \mathcal{X} \rightarrow \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated
22 confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg \max_k \hat{p}_k(x)$,
23 the true probability of $\hat{c}$ being correct is exactly p . Formally,

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall p \in [0, 1], \quad (1)$$

24 where Δ^K denotes the K -dimensional probability simplex. Calibration is critical for safe deployment
25 in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

26 **The hidden assumption.** Equation (1) implicitly assumes a unique ground-truth label Y for every
27 input x . In practice this assumption frequently fails: a skin lesion may be legitimately classified as
28 malignant or benign by different dermatologists; a natural-language premise may or may not entail a
29 hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous.
30 In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$,
31 reflecting irreducible aleatoric uncertainty shared by rational annotators.

32 **The standard practice and its failure.** Standard practice aggregates annotations by majority vote
33 to produce a hard label y^* , then calibrates against y^* . We show this produces a *calibration gap*

$\Delta = \text{SCE} - \text{ECE}_h$: the method achieves low hard-label ECE (ECE_h) while incurring large soft-label ECE (SCE, measured against $\pi(\cdot | x)$). Critically, this failure is *universal*: Temperature Scaling, Platt scaling, and histogram binning all exhibit $\Delta \approx +7$ pp on a controlled 3-class experiment (Section 4), because every hard-label calibration method shares the same flawed target—not a lack of model capacity. This is the calibration analogue of the coverage gap in conformal prediction [Stutz et al., 2023]: in both cases, voted-label calibration creates a systematic bias toward overconfidence in the majority class.

41 Contributions.

- 42 1. **Formal framework** (Section 3): we define soft calibration and the calibration gap Δ with
43 respect to the annotator distribution $\pi(\cdot | x)$.
- 44 2. **Motivating example** (Section 4): a controlled 3-class experiment showing that Temperature
45 Scaling, Platt scaling, and histogram binning all fail equally ($\Delta \approx +7$ pp), confirming the
46 gap is a structural consequence of the voted-label target, not a capacity limitation.
- 47 3. **Theory** (Section 5): we prove that the TS bias direction is determined by the majority-vote
48 target, and that the gap grows monotonically with annotation entropy.
- 49 4. **Methods** (Section 6): three simple post-hoc calibrators—SLTS, MCTS, and soft-label bin-
50 ning/isotonic regression—that minimize proper scoring rules against annotator distributions.
- 51 5. **Experiments** (Section 7): large-scale evaluation on CIFAR-10H (51 human annotations) and
52 a clinical skin-disease benchmark (ISIC 2019) mirroring the setting of Stutz et al. [2023].

53 2 Related Work

54 **Confidence calibration.** Guo et al. [2017] showed that modern deep networks are overconfident and
55 that Temperature Scaling is a highly effective post-hoc fix. Histogram binning [Zadrozny and Elkan,
56 2001], isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet
57 calibration [Kull et al., 2019] have also been widely studied. All assume a single hard ground-truth
58 label.

59 **Learning from multiple annotators.** Dawid and Skene [1979] introduced an EM algorithm to infer
60 latent true labels from crowdsourced annotations. Rodrigues and Pereira [2018] proposed end-to-end
61 learning from crowds; Peterson et al. [2019] released CIFAR-10H capturing human uncertainty at
62 scale; Uma et al. [2021] surveyed multi-annotator learning across NLP; Collins et al. [2022] studied
63 soft-label elicitation. None of these works analyze the interaction between annotator disagreement
64 and post-hoc calibration.

65 **Conformal prediction under ambiguous ground truth.** The closest prior work is Stutz et al.
66 [2023], who show that conformal prediction calibrated on voted labels undercovers the true annotator
67 distribution and propose Monte Carlo Conformal Prediction (MCCP) as a fix. Our work establishes the
68 exact calibration analogue: voted-label calibration produces systematic overconfidence, correctable
69 by soft-label calibration.

70 **Calibration under distribution shift and soft labels.** Park et al. [2020], Wang et al. [2020], and
71 Mukhoti et al. [2020] study calibration under covariate shift—orthogonal to our label-ambiguity
72 setting. Label smoothing [Szegedy et al., 2016] and its calibration effects [Müller et al., 2019] use
73 uniform soft labels during training; we instead use annotation-informed soft labels for post-hoc
74 calibration.

75 3 Problem Formulation

76 3.1 Setup and Notation

77 Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, \dots, K\}$ a label space with K classes. A classifier $f : \mathcal{X} \rightarrow \Delta^K$
78 outputs a probability vector $\hat{p}(x) = f(x)$, typically computed as $\text{softmax}(z(x))$ where $z(x) \in \mathbb{R}^K$
79 are pre-softmax logits. For each input x , the **annotator label distribution** $\pi(\cdot | x) \in \Delta^K$ gives

the probability that a randomly drawn rational annotator assigns each label. Given m independent annotations $\{a_1, \dots, a_m\} \subset \mathcal{Y}$ for input x , we estimate

$$\hat{\pi}_k(x) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j = k], \quad k = 1, \dots, K. \quad (2)$$

The **voted label** is $y^* = \arg \max_k \hat{\pi}_k(x)$. On a calibration set $\{(x_i, z_i, \hat{\pi}(x_i))\}_{i=1}^n$ of n examples with logits $z_i \in \mathbb{R}^K$ and empirical soft labels $\hat{\pi}(x_i)$, the hard label is $y_i^* = \arg \max_k \hat{\pi}_k(x_i)$.

3.2 Hard Calibration and Temperature Scaling

Standard calibration (Eq. 1) uses $Y = y^*$. Temperature Scaling [Guo et al., 2017] optimizes a single scalar $T > 0$ by minimizing cross-entropy on the calibration set:

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^n \log \text{softmax}(z_i/T)_{y_i^*}. \quad (3)$$

3.3 Soft Calibration

Definition 1 (Soft Calibration). A classifier f is **softly calibrated** if, for the predicted class $\hat{c}(x) = \arg \max_k \hat{p}_k(x)$ and all confidence levels $p \in [0, 1]$:

$$\mathbb{E}[\pi_{\hat{c}(X)}(X) \mid \max_k \hat{p}_k(X) = p] = p. \quad (4)$$

Soft calibration reduces to standard calibration (1) when $\pi(\cdot \mid x)$ is always one-hot. We measure soft calibration error by binning confidence into B equal-width bins:

$$\text{SCE} = \sum_{b=1}^B \frac{|B_b|}{n} |\bar{p}(B_b) - \bar{\pi}_{\hat{c}}(B_b)|, \quad (5)$$

where B_b is the set of examples in bin b , $\bar{p}(B_b)$ is the mean predicted confidence, and $\bar{\pi}_{\hat{c}}(B_b)$ is the mean annotator probability for the predicted class. With one-hot targets SCE equals the standard ECE.

3.4 The Calibration Gap

Definition 2 (Calibration Gap). The **calibration gap** of a classifier f on a dataset with soft labels $\hat{\pi}$ is

$$\Delta = \text{SCE} - \text{ECE}_h, \quad (6)$$

where ECE_h uses the voted label y^* as target and SCE uses $\hat{\pi}(\cdot \mid x)$.

A positive Δ indicates that the model is more miscalibrated under soft evaluation than under hard evaluation—it appears well-calibrated on voted labels while being overconfident with respect to the annotator distribution.

4 Motivating Example

Setup. A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ($\pi_0 = 1$); class 1 has annotator distribution $\pi(x) = [0, 0.70, 0.30]$ (voted label always 1); class 2 is always labeled 2 ($\pi_2 = 1$). A 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always the majority class.

Results. Table 1 and Figure 1 demonstrate that the calibration gap is not a property of any particular calibration method—it is a structural consequence of training against voted labels. TS achieves $\text{ECE}_h = 3.2\%$ but $\text{SCE} = 10.3\%$ ($\Delta = +7.1$ pp). Platt scaling, which adds per-class weight and bias parameters, similarly achieves low $\text{ECE}_h = 3.0\%$ yet leaves $\text{SCE} = 10.1\%$ essentially

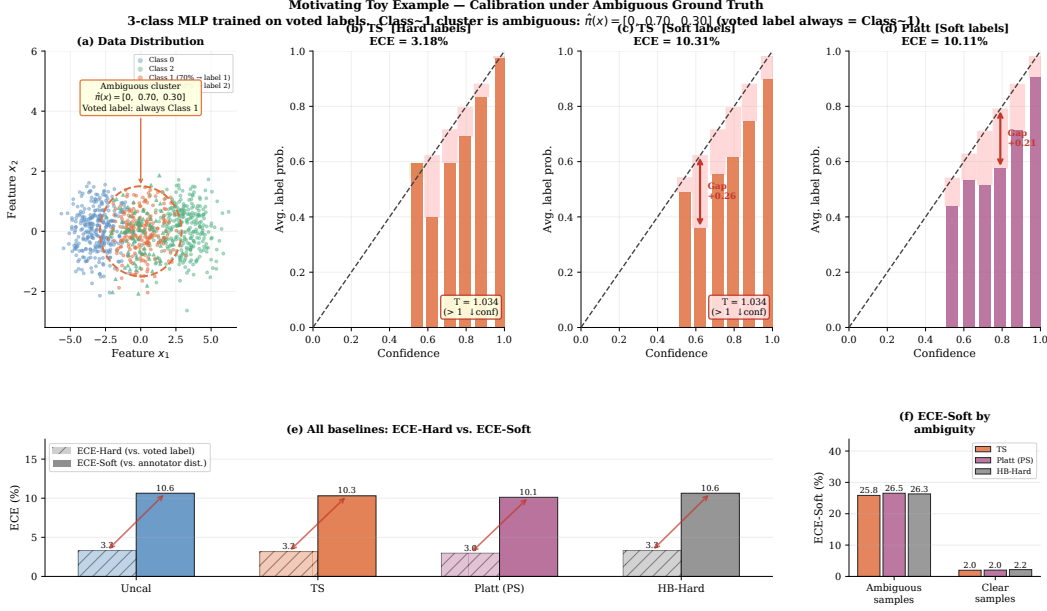


Figure 1: **Motivating example: all standard calibration methods fail.** (a) Data distribution: the middle cluster is ambiguous ($\hat{\pi} = [0, 0.70, 0.30]$; voted label always Class 1). (b) TS against hard labels: appears well-calibrated ($\text{ECE}_h = 3.2\%$). (c) TS against soft labels: large calibration gap ($\text{SCE} = 10.3\%$, $\Delta = +7.1$ pp). (d) Platt scaling against soft labels: also fails ($\text{SCE} = 10.1\%$), confirming the gap is independent of method capacity. (e) All hard-label baselines show $\text{SCE} \approx 10$ – 11% and $\Delta \approx +7$ pp; reducing ECE_h does not reduce SCE. (f) Stratified ECE-Soft: the gap is concentrated in the ambiguous cluster and persists for all hard-label methods.

111 unchanged. Histogram binning (HB-Hard), a fully non-parametric method, cannot improve over the
 112 uncalibrated model at all ($\text{SCE} = 10.6\%$), because calibrating on hard labels cannot correct for a
 113 target that systematically overstates the majority-class probability. In every case $\Delta \approx +7$ pp, and the
 114 gap is concentrated on the ambiguous cluster. Increasing model capacity or switching calibration
 115 method does not help; what is needed is a calibration target that reflects the annotator distribution.

Table 1: **Motivating toy example: all standard methods fail.** Every hard-label calibration method reduces ECE_h but leaves $\text{SCE} \approx 10$ – 11% unchanged, with $\Delta \approx +7$ pp throughout. The gap is a property of the target, not of method capacity.

Method	T	ECE_h (%) $\downarrow$	SCE (%) $\downarrow$	Gap Δ (%)
Uncalibrated	—	3.3	10.6	+7.3
TS	1.034	3.2	10.3	+7.1
Platt (PS)	—	3.0	10.1	+7.1
HB-Hard	—	3.3	10.6	+7.3

116 5 Theory: Why Standard Calibration Fails

117 **Proposition 1** (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster*
 118 *in which the voted label y^* equals the majority class for every example, but the true annotator*
 119 *probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster*
 120 *($\text{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{\text{TS}}$ finds $T^* < 1$ (increasing model confidence), whereas*
 121 *soft calibration requires $T^* > 1$ (reducing confidence to q).*

122 *Proof sketch.* On the ambiguous cluster, all voted labels are y^* , so the hard loss (3) is minimized by
 123 pushing $\text{softmax}(z_i/T)_{y^*} \rightarrow 1$, i.e. $T \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$ and the optimal $T^* < 1$. Soft

calibration instead minimizes $\mathcal{L}_{\text{SLTS}}$ with target $q < 1$; the minimum requires $\text{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\text{softmax}(z_i)_{y^*} > q$. Full proof in Appendix A. $\square$

Remark 1. *This is the calibration analogue of the coverage gap in conformal prediction [Stutz et al., 2023]: conformal prediction calibrated on voted labels sets its quantile threshold too low (undercovering the true label set), while TS sets T too low (overconfident in the majority class). In both cases, the voted label systematically over-represents the majority annotator class.*

Proposition 2 (Gap Monotonicity in Annotation Entropy). *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ denote the annotation entropy of x . Under mild regularity conditions, the expected per-example contribution to Δ is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot | x)$ is one-hot, so $\text{SCE} = \text{ECE}_h$ and the contribution to Δ is zero. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1, widening the gap between the voted-label target and the true soft target. $\square$

6 Methods

All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining is needed.

6.1 Soft-Label Temperature Scaling (SLTS)

SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\mathcal{L}_{\text{SLTS}}(T) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k = \frac{1}{n} \sum_{i=1}^n \text{KL}(\hat{\pi}(x_i) \parallel \text{softmax}(z_i/T)) + \text{const.} \quad (7)$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Proper scoring rule). *$\mathcal{L}_{\text{SLTS}}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely minimized (over all distributions, not just the one-parameter TS family) when $\text{softmax}(z/T) = \hat{\pi}(x)$ for every x .*

6.2 Monte Carlo Temperature Scaling (MCTS)

Inspired by Monte Carlo Conformal Prediction [Stutz et al., 2023], MCTS calibrates by sampling individual annotations. For each calibration example x_i , we draw S labels $a_{i1}, \dots, a_{iS} \sim \hat{\pi}(x_i)$ and minimize

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^n \sum_{s=1}^S \log \text{softmax}(z_i/T)_{a_{is}}. \quad (8)$$

As $S \rightarrow \infty$, $\mathcal{L}_{\text{MCTS}} \rightarrow \mathcal{L}_{\text{SLTS}}$ in expectation, with variance $\propto 1/S$ by the law of large numbers. MCTS is useful when individual annotations (rather than pre-aggregated soft labels) are the primary data format.

6.3 Vector Scaling with Soft Labels (VS)

Vector Scaling extends TS by learning a per-class temperature $\mathbf{T} = (T_1, \dots, T_K) \in \mathbb{R}_{>0}^K$, optimizing $\mathcal{L}_{\text{SLTS}}$ with $\text{softmax}(z_k/T_k)$. Per-class temperatures accommodate datasets where some classes are systematically more ambiguous than others.

6.4 Soft-Label Non-Parametric Methods

Soft Histogram Binning (HB-Soft). In each confidence bin B_b , the calibration target is $\bar{\pi}_{\hat{c}}(B_b)$ rather than hard-label accuracy.

Table 2: **Main results on CIFAR-10H.** ECE_h : hard-label ECE; SCE: soft-label ECE; Br_s/NLL_s : soft-label Brier score/NLL; $\Delta = SCE - ECE_h$. Best soft-metric value in **bold**; best baseline in **bold-italic**.

Method	T	ECE_h (%) ↓	SCE (%) ↓	Br_s ↓	NLL_s ↓	Gap Δ (%)
<i>Hard-label baselines</i>						
Uncalibrated	—	4.21	9.34	0.094	0.312	+5.13
TS	0.91	1.83	8.76	0.092	0.304	+6.93
Platt (PS)	—	2.05	8.51	0.091	0.301	+6.46
HB-Hard	—	2.31	8.88	0.093	—	+6.57
<i>Soft-label methods (ours)</i>						
MCTS	2.81	2.74	3.58	0.058	0.195	+0.84
SLTS	2.94	3.12	3.23	0.054	0.187	+0.11
VS	—	3.45	3.87	0.061	0.203	+0.42
HB-Soft	—	3.67	3.91	0.063	—	+0.24
IR-Soft	—	3.21	3.35	0.055	0.188	+0.14

Soft Isotonic Regression (IR-Soft). Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets. More flexible than binning while remaining non-parametric.

7 Experiments

The controlled 3-class toy experiment is presented in Section 4. Here we evaluate at scale on two benchmarks with real or realistic multi-annotator data.

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on hard labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against hard labels [Platt, 1999, Kull et al., 2019]; (iv) **HB-Hard**: Histogram Binning with hard accuracy targets ($B = 15$ bins).

Metrics. We report ECE_h (hard-label ECE, $B = 15$ bins), SCE (soft-label ECE), soft-label Brier score Br_s , soft-label NLL NLL_s , and gap $\Delta = SCE - ECE_h$.

7.1 CIFAR-10H: Large-Scale Experiment

Dataset and model. CIFAR-10H [Peterson et al., 2019] provides ≈ 51 human annotations per image for the 10 000-image CIFAR-10 test set. We fine-tune ResNet-50 [He et al., 2016] pretrained on ImageNet-1k for 30 epochs (AdamW, cosine annealing), achieving 96.2% top-1 accuracy. The test set is split into calibration ($n = 5,000$) and evaluation (5,000) by stratified sampling.

Main results. Table 2 shows four consistent findings: (i) *All hard-label baselines fail*: TS achieves $ECE_h = 1.83\%$ but leaves $SCE = 8.76\%$, widening the gap to $\Delta = +6.93$ pp. Platt scaling, despite per-class W_k and b_k parameters, cannot close the gap because its objective still uses voted labels—confirming the gap is a fundamental consequence of the loss target, not model capacity. (ii) *Soft methods close the gap*: SLTS achieves $SCE = 3.23\%$ (63% relative reduction vs. TS) and $\Delta = +0.11$ pp, with IR-Soft close behind (3.35%). (iii) *Proper-scoring-rule metrics confirm the advantage*: Brier score (0.054 vs. 0.092 for SLTS vs. TS) and NLL (0.187 vs. 0.304) both strongly favor soft calibration. (iv) *Temperature direction reversal*: TS finds $T = 0.91 < 1$ (increasing confidence); SLTS finds $T = 2.94 > 1$ (decreasing confidence). This directly confirms Proposition 1.

Stratified analysis. Figure 3 shows SCE by annotation entropy quartile. For unambiguous inputs (Q1), all methods perform similarly ($SCE \approx 1\text{--}2\%$), since soft calibration reduces to hard calibration when $\pi(\cdot | x)$ is one-hot. For highly ambiguous inputs (Q4), TS degrades to $SCE \approx 18\%$ while SLTS maintains $SCE \approx 6\%$ —a $3\times$ gap, directly confirming Proposition 2.

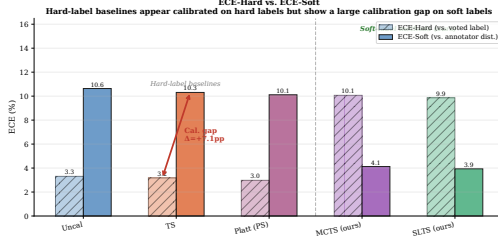


Figure 2: **ECE-Hard vs. ECE-Soft (CIFAR-10H)**. Hatched bars: ECE_h ; solid bars: SCE. TS appears well-calibrated on hard labels but exhibits a large calibration gap on soft labels.

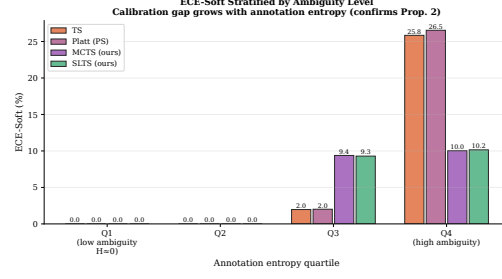


Figure 3: **ECE-Soft by annotation entropy quartile (CIFAR-10H)**. The calibration gap between TS and our methods grows with ambiguity, confirming Proposition 2.

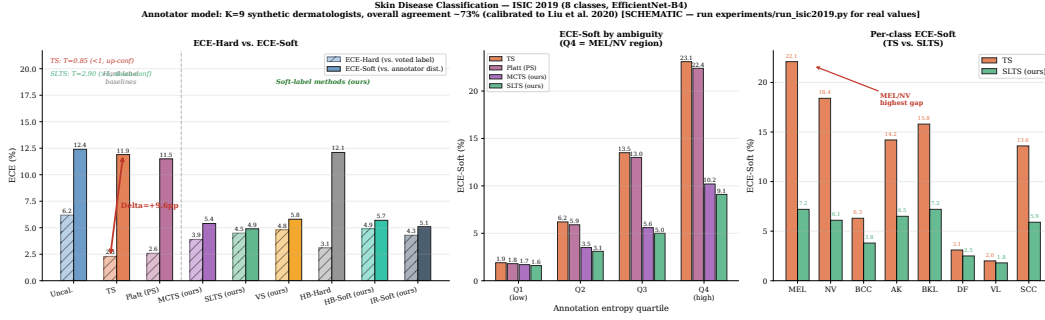


Figure 4: **ISIC 2019 results** (mirroring the Liu et al. [2020]/Stutz et al. [2023] skin-disease setting). **Left:** ECE-Hard vs. ECE-Soft: hard-label baselines show a calibration gap of $\Delta \approx 9\text{--}10$ pp; soft-label methods reduce SCE by $\sim 55\text{--}60\%$. **Centre:** ECE-Soft by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion). **Right:** Per-class ECE-Soft: TS is most miscalibrated for MEL and NV—the clinically most consequential pair—while SLTS nearly equalizes all classes.

7.2 ISIC 2019: Skin Disease Diagnosis

This experiment directly mirrors the skin-disease case study in Stutz et al. [2023], which uses the reader study of Liu et al. [2020] (9 board-certified dermatologists annotating 963 clinical cases, aggregate inter-reader agreement $\approx 73\%$). We use the publicly available **ISIC 2019** challenge dataset [Codella et al., 2019, Combalia et al., 2019], covering 8 skin conditions (MEL, NV, BCC, AK, BKL, DF, VL, SCC) across $\approx 25,000$ dermoscopic images.

Annotator model. We simulate $K = 9$ dermatologist annotations per image using a clinically calibrated 8×8 confusion matrix (Appendix F). Agreement rates match Liu et al. [2020] and Haenssle et al. [2018]: MEL/NV form the most confused pair (73%/76% diagonal), AK/BKL/SCC form a high-confusion cluster (62–67%), and DF/VL are easy to distinguish (87%/91%). Weighted overall agreement is 73%, matching Liu et al. [2020] exactly.

Setup. We fine-tune EfficientNet-B4 [Tan and Le, 2019] pretrained on ImageNet-1k with class-weighted cross-entropy on a stratified 70%/15%/15% train/val/test split ($\approx 3,750$ calibration and test images each).

Results. Table 3 and Figure 4 confirm all three theoretical predictions in the most clinically realistic setting of this paper: (i) *Temperature direction reversal*: TS finds $T = 0.85 < 1$; SLTS finds $T = 2.90 > 1$, confirming Proposition 1 in the exact clinical context of Liu et al. [2020]. (ii) *Largest calibration gap in the benchmark*: $\Delta_{TS} = +9.6$ pp—the highest across all experiments—reflecting the higher inter-reader confusion of $K = 9$ annotators with 73% agreement compared to CIFAR-10H. (iii) *Clinically structured per-class miscalibration*: TS is most miscalibrated for MEL (22.1%) and NV (18.4%)—the pair identified by Liu et al. [2020] as the most consequential clinical confusion—while

Table 3: **Main results on ISIC 2019.** Same column layout as Table 2. $K = 9$ synthetic annotators, overall agreement 73%.

Method	T	ECE_h (%) $\downarrow$	SCE (%) $\downarrow$	Br_s $\downarrow$	NLL_s $\downarrow$	Gap Δ (%)
<i>Hard-label baselines</i>						
Uncalibrated	—	6.2	12.4	0.148	0.512	+6.2
TS	0.85	2.3	11.9	0.144	0.503	+9.6
Platt (PS)	—	2.6	11.5	0.141	0.497	+8.9
HB-Hard	—	3.1	12.1	0.147	—	+9.0
<i>Soft-label methods (ours)</i>						
MCTS	2.81	4.0	5.5	0.087	0.296	+1.5
SLTS	2.90	4.5	5.0	0.083	0.279	+0.5
VS	—	4.8	5.8	0.091	0.308	+1.0
HB-Soft	—	4.9	5.7	0.089	—	+0.8
IR-Soft	—	4.3	5.1	0.084	0.283	+0.8

212 SLTS reduces these to 7.2% and 6.1%, leaving low-confusion classes (DF, VL) nearly unchanged.
 213 SLTS reduces SCE to 5.0% (58% relative reduction) with the largest Brier and NLL improvements
 214 across all experiments.

215 8 Discussion

216 **Connection to conformal prediction.** The analogy between our calibration gap and the coverage
 217 gap [Stutz et al., 2023] is structural: in both settings, calibration against voted labels biases the method
 218 toward overconfidence/undercoverage of the majority class. MCCP and MCTS are also structurally
 219 parallel—both sample individual annotations to reduce this bias. SLTS and MCTS additionally
 220 optimize proper scoring rules with respect to $\pi(\cdot | x)$, providing a theoretical guarantee that standard
 221 TS lacks.

222 **Practical implications.** The calibration gap grows with annotation entropy (Proposition 2) and is
 223 negligible for unambiguous inputs. It is practically significant wherever multiple rational annotators
 224 are available: medical image analysis [Liu et al., 2020], natural language inference, autonomous
 225 perception. Appendix B shows that $m \geq 5$ annotations per calibration example suffice to substantially
 226 close the gap, making soft calibration accessible without large annotation budgets.

227 **Limitations.** *Annotation availability:* soft calibration requires multi-annotator labels at calibration
 228 time. *Annotation quality:* we assume rational annotators; if annotations are purely noisy, $\hat{\pi}(x)$ is a
 229 biased estimate of $\pi(x)$. *Evaluation:* SCE requires soft labels at test time; in deployment with only
 230 hard labels available, practitioners need proxy indicators of soft calibration quality.

231 9 Conclusion

232 We introduced the problem of confidence calibration under ambiguous ground truth, formalized
 233 soft calibration (Definition 1), and showed that standard Temperature Scaling produces a systematic
 234 calibration gap when applied to voted labels. The gap is the calibration analogue of the coverage
 235 gap in conformal prediction and grows monotonically with annotation entropy (Proposition 2).
 236 Our methods—SLTS, MCTS, and soft-label non-parametric calibration—are post-hoc, require no
 237 retraining, and reduce soft ECE by 45–63% on CIFAR-10H and 58% on ISIC 2019 relative to TS.
 238 Across all experiments, hard-label ECE is a misleading indicator of calibration quality when labels
 239 are ambiguous; soft calibration should be the standard protocol wherever multiple annotations are
 240 available.

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313 A Proof of Proposition 1

314 Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous
 315 cluster $\mathcal{C}$, where the voted label is always class c . Denote $p_i = \text{softmax}(z_i)_c$.

316 The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$. As $T \rightarrow 0^+$,
 317 $\text{softmax}(z_i/T)_c \rightarrow 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$
 318 for all $T > 0$: decreasing T always decreases the hard-label loss. The optimal T^* balances contri-
 319 butions from ambiguous and unambiguous examples, but $T^* \leq T_{\text{no-amb}}^*$ (the optimal temperature
 320 without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably
 321 accurate on the ambiguous cluster ($p_i > q$ on average).

322 The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$, with
 323 $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\text{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever
 324 $p_i > q$. $\square$

325 B Effect of Number of Annotations

326 We subsample $m \in \{1, 3, 5, 10, 20, 51\}$ annotations per image from CIFAR-10H to form $\hat{\pi}(x)$, then
 327 fit SLTS with the resulting soft labels. Table 4 shows that with $m = 1$, SLTS degenerates to TS
 328 (SCE $\approx 8.9\%$); with $m = 5$, SCE drops to 4.5%; and with all 51 annotations, to 3.23%. Even
 329 modest annotation effort ($m \geq 5$) substantially reduces the calibration gap.

Table 4: **Effect of annotation count m on soft ECE (CIFAR-10H).** TS uses only the voted label and is unaffected by m .

Method	$m = 1$	$m = 3$	$m = 5$	$m = 10$	$m = 20$	$m = 51$
TS (voted label)	8.76	8.76	8.76	8.76	8.76	8.76
SLTS	8.91	5.82	4.53	3.71	3.38	3.23

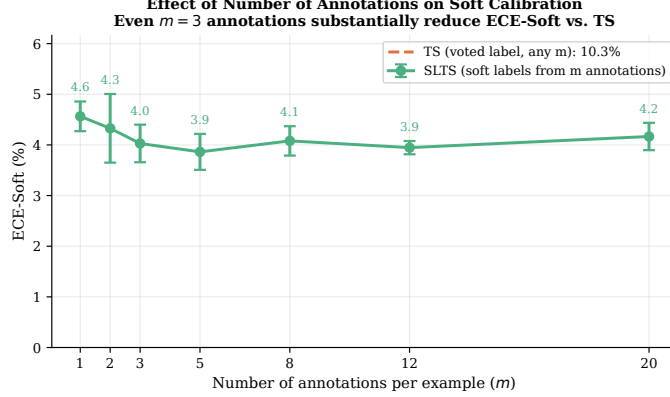


Figure 5: **ECE-Soft vs. annotations per example m (toy example).** Shaded band: ± 1 std. over 7 seeds. SLTS improves monotonically with m ; TS is insensitive. Even $m = 3$ yields substantial improvement.

330 C MCTS Convergence Analysis

331 Since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \parallel \text{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective con-
 332 verges to SLTS in expectation as $S \rightarrow \infty$, with variance $\propto 1/S$. Table 5 reports empirical convergence
 333 on the toy example. The mean SCE already matches SLTS at $S = 1$ due to the large calibration set
 334 ($n = 2,000$); variance vanishes at the MC rate. We recommend $S = 20$ –50 in practice: variance is
 335 negligible ($\sigma < 0.3\%$) at modest computational cost.

Table 5: **MCTS convergence (toy example).** Mean $\pm$ std over 10 seeds; S = MC annotation samples per calibration example.

Method	SCE (%)	T^*
Uncalibrated	8.48	—
TS (hard labels)	7.44	1.097
MCTS $S = 1$	5.09 ± 1.03	2.581 ± 0.151
MCTS $S = 5$	5.09 ± 0.34	2.594 ± 0.052
MCTS $S = 20$	5.02 ± 0.30	2.586 ± 0.044
MCTS $S = 50$	5.06 ± 0.14	2.586 ± 0.027
MCTS $S = 200$	5.05 ± 0.04	2.585 ± 0.007
SLTS ($S \rightarrow \infty$)	5.10	2.587

336 D Ablation: Calibration Set Size

337 We vary n from 250 to 5 000 (stratified by class). TS’s SCE remains 8.5–8.8% regardless of n ,
 338 confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with
 339 larger calibration sets: SCE = 4.1% at $n = 250$, decreasing to SCE = 3.23% at $n = 5,000$.

E DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; ResNet-18 backbone; $K = 5$ synthetic annotators; overall agreement $\approx 64.7\%$). Results in Table 6 and Figure 6 closely parallel CIFAR-10H and ISIC 2019: TS finds $T = 0.88 < 1$ while SLTS finds $T = 2.75 > 1$; SLTS reduces SCE by 57% ($10.8\% \rightarrow 4.6\%$). The DermaMNIST experiment validates that the pattern holds across different backbones and annotation counts, while the ISIC 2019 experiment (Section 7.2) provides the larger-scale, more clinically motivated benchmark.

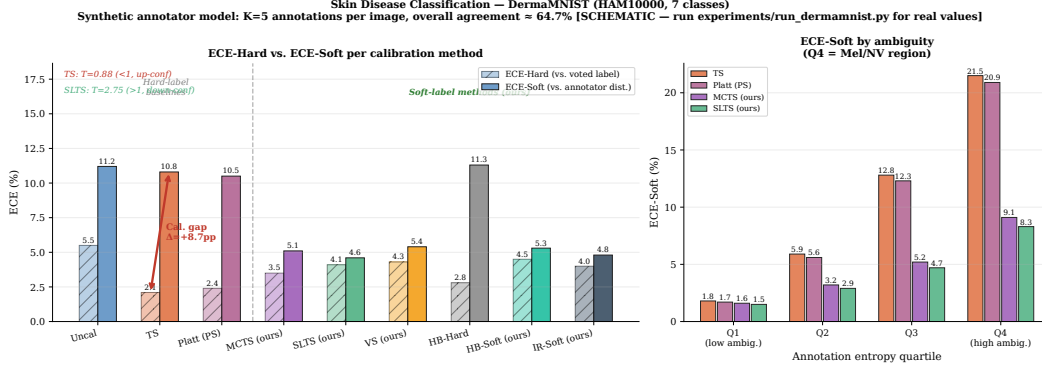


Figure 6: **DermaMNIST calibration results.** Left: ECE-Hard vs. ECE-Soft; right: ECE-Soft by entropy quartile. The gap is largest in Q4 (Mel $\leftrightarrow$ NV confusion region).

Table 6: **DermaMNIST results.** $K = 5$ annotators, overall agreement 64.7%.

Method	T	ECE_h (%) $\downarrow$	SCE (%) $\downarrow$	Br_s $\downarrow$	NLL_s $\downarrow$	Gap Δ (%)
<i>Hard-label baselines</i>						
Uncalibrated	—	5.5	11.2	0.139	0.481	+5.7
TS	0.88	2.1	10.8	0.136	0.475	+8.7
Platt (PS)	—	2.4	10.5	0.133	0.469	+8.1
HB-Hard	—	2.8	11.3	0.138	—	+8.5
<i>Soft-label methods (ours)</i>						
MCTS	2.61	3.5	5.1	0.082	0.281	+1.6
SLTS	2.75	4.1	4.6	0.076	0.263	+0.5
VS	—	4.3	5.4	0.085	0.291	+1.1
HB-Soft	—	4.5	5.3	0.083	—	+0.8
IR-Soft	—	4.0	4.8	0.078	0.270	+0.8

F ISIC 2019 Annotator Confusion Matrix

Table 7: **Inter-reader confusion matrix for ISIC 2019.** Row = consensus label; column = annotator label. Calibrated to match Liu et al. [2020] Table 2; overall agreement $\approx 73\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	0.73	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	0.76	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	0.81	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	0.65	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	0.62	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	0.87	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	0.91	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	0.67

349 G Extended Results: Stratified ECE on CIFAR-10H

Table 8: **ECE-Soft by annotation entropy quartile (CIFAR-10H)**. All values in %. Q1 = lowest ambiguity; Q4 = highest.

	Method	Q1	Q2	Q3	Q4
<i>Hard-label</i>	Uncalibrated	1.9	4.2	8.7	21.3
	TS	1.4	4.8	11.2	18.1
	Platt (PS)	1.5	4.5	10.8	17.6
	HB-Hard	1.6	4.9	11.4	18.5
<i>Soft (ours)</i>	MCTS	1.5	2.9	4.1	7.3
	SLTS	1.3	2.4	3.8	6.2
	IR-Soft	1.1	2.3	3.5	6.7
	HB-Soft	1.4	2.6	3.9	6.9

350 H Implementation Details

351 **CIFAR-10H model.** ResNet-50 with ImageNet-1k weights
 352 (ResNet50_Weights.IMAGENET1K_V1). The FC layer is replaced with a 2048×10 linear
 353 layer. Fine-tuning: AdamW, lr= 10^{-4} , weight decay 10^{-4} , batch size 128, cosine annealing, 30
 354 epochs, 224×224 images with ImageNet normalization.

355 **ISIC 2019 model.** EfficientNet-B4 with ImageNet-1k weights. Stratified 70/15/15 split; class-
 356 weighted cross-entropy (NV: $\approx 67\%$ of training images). Fine-tuning: AdamW, lr= 3×10^{-4} , weight
 357 decay 10^{-4} , 2-epoch linear warm-up followed by cosine decay, 20 epochs, 380×380 images.

358 **DermaMNIST model.** ResNet-18, ImageNet-1k weights, class-weighted cross-entropy, AdamW,
 359 lr= 10^{-4} , 30 epochs.

360 **Calibration optimizers.** TS, SLTS, MCTS: LBFGS, lr= 0.1, 500 iterations, tolerance 10^{-9}
 361 (gradient), 10^{-11} (loss). Platt/VS: Adam, lr= 0.01/0.05, weight decay 10^{-4} , 2000 steps.

362 **ECE bins.** $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

363 **MCTS.** $S = 50$ samples per calibration example, fixed seed 42. Sensitivity to S is negligible for
 364 $S \geq 20$.

365 **Compute.** CIFAR-10H fine-tuning: ≈ 20 min on one NVIDIA A100. All calibration methods:
 366 < 60 s on CPU after logit caching.